

Evaluating Human-AI Privacy Alignment: Interactive Teaching of Contextual Privacy Preferences

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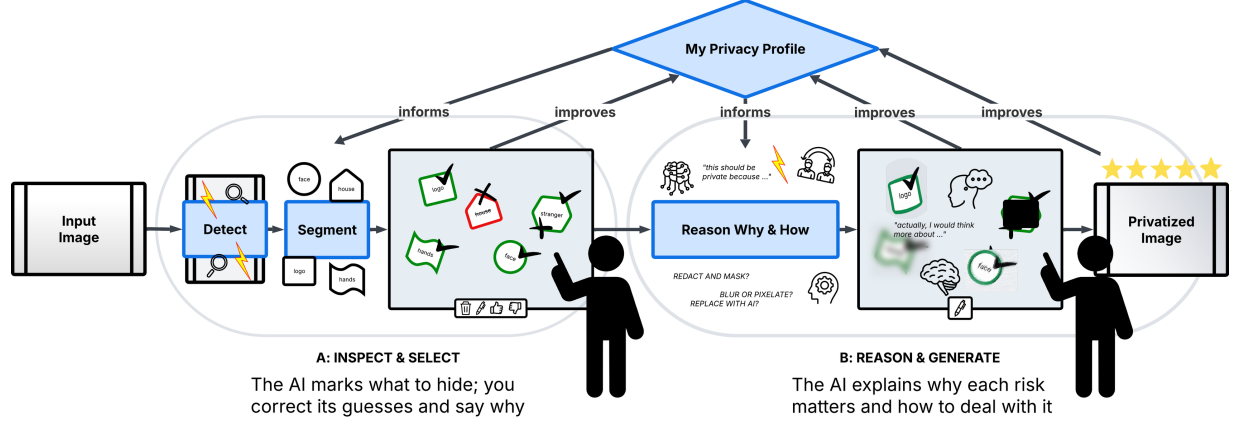


Figure 1: The mixed-initiative, co-adaptive teaching workflow instantiated by Pringl. Each image passes through Stage A (*Inspect & Select*), where the user corrects proposed regions, and Stage B (*Reason & Generate*), where the user reviews per-region rationales (“whys”) and edit actions (“hows”). Every correction informs a central Privacy Profile, which in turn improves both stages on later runs.

ABSTRACT

Privacy in personal images is not a fixed property that automated anonymization can optimize for: it is a personal, context-dependent judgment. This paper asks two questions: *How well can AI grasp a user’s contextual privacy preferences?* (RQ1) and *Can we teach and align AI to how humans reason about privacy?* (RQ2). We propose a mixed-initiative, co-adaptive workflow in which the AI takes the first pass at privatizing an image and the user steers it from there by keeping, fixing, or adding regions and explaining *why*, until both agree on what private *means*. Accepted corrections and their rationales accumulate in an inspectable, reusable Privacy Profile that guides later runs. We implement this workflow in Pringl, a web prototype, and report qualitative sessions in which learned profiles transfer to unseen images; when they do not, the error usually reflects a gap in what was taught.

Index Terms: Anonymization, image privacy, privacy-preserving machine learning, personalized privacy assistant, user-centric privacy.

1 INTRODUCTION

AI developers need human data to improve their models, but that data can leak sensitive information: training images can be extracted from generative models, and inputs can be reconstructed from model outputs or shared gradients [5, 7, 24]. Privatizing images by hand does not scale, and standard automated remedies such as scrubbing identifiers or adding differentially private noise during training [6, 21] can *over-privatize* and reduce utility, destroying useful image qualities, or *under-privatize* and leave exploitable

cues. Blurred and pixelated content can be recovered [14, 27], and people remain recognizable even with faces hidden [18].

The deeper problem is that human-centered privacy is not fixed. HCI research treats it as a situated practice: Nissenbaum’s contextual integrity ties appropriateness to the actors, attributes, and transmission norms of a context [17], and Palen and Dourish describe privacy as boundary negotiation across audiences and contexts [19]. The same visual cue may be acceptable for one audience yet sensitive for another, and people judge the same cue differently even under similar conditions [1, 13]. Because such judgments lack a single ground truth, one-shot automatic anonymization is a poor fit; interactive machine learning instead suggests mixed-initiative workflows in which users inspect concrete outputs, correct proposals, and steer a personalized policy over time [2]. Before such agents could act on a person’s behalf, researchers need a way to test whether alignment to individual preferences is feasible at all. We therefore ask:

RQ1: How well can AI grasp a user’s contextual privacy preferences?

RQ2: Can we teach and align AI to how humans reason about privacy?

We make three contributions, deliberately separating the general pipeline from our implementation of it. First, we introduce the *Privacy Profile*, an inspectable representation of evolving privacy preferences—a tree of categories, learned concepts, grounding examples, and edit policies rather than an opaque policy dump (Sec. 3.3). Second, we propose a system-agnostic, mixed-initiative, co-adaptive teaching pipeline that converts region-level corrections and their rationales into such a profile (Sec. 3, Fig. 1). Third, we implement this pipeline in Pringl (Personal, Readable, Interactive, Negotiated, Grounded, Local), a web prototype supporting teaching on reference images and read-only evaluation on unseen ones (Sec. 4), and report qualitative transfer results (Sec. 5).

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2 RELATED WORK

2.1 Contextual and Interactive Image Privacy

Most learned anonymization optimizes a fixed objective, such as fooling a recognizer while preserving task utility [23, 26], leaving no room for individual judgment. Closer to the contextual view of Sec. 1, Imago Obscura [16] is our main point of departure: an image-privacy AI copilot that helps users articulate sharing intent, surface contextually relevant risks, and apply mitigations such as blurring, inpainting, and generative replacement. Pringl adopts this interaction-centered view but changes the unit of analysis: rather than helping a user edit one image, every interaction becomes training signal for a reusable, per-user Privacy Profile. The approach is closer in spirit to learning privacy norms from people directly [25], but applied to concrete images instead of survey vignettes.

2.2 Interactive ML and User Modeling

Interactive machine learning studies loops in which users iteratively correct model behavior and the system adapts in turn [2]. Preference learning at scale typically follows reinforcement learning from human feedback, which distills comparisons from many annotators into one population-level reward model [11]; cooperative inverse reinforcement learning likewise frames alignment as inferring what a human values from their behavior [9]. Our setting differs in two ways. The preferences are individual rather than population-level, and the learned user model is not hidden in weights: the Privacy Profile is an explicit, human-readable artifact that both the modeled user and a researcher studying them can inspect, audit, and correct. This matters for privacy mechanisms in particular, which must remain understandable to lay users to be trusted [28, 20, 10].

3 THE MIXED-INITIATIVE, CO-ADAPTIVE TEACHING WORKFLOW

We first describe the pipeline in system-agnostic terms; Sec. 4 describes Pringl, our implementation of it. Rather than asking users to specify a complete privacy policy upfront, the workflow lets them teach from examples: images, corrections, and justifications accumulate in a reusable *Privacy Profile*.

3.1 Design Goals

DG1: Support contextual preference expression. Preferences often surface only when users face a concrete image and use case, so the workflow never asks for abstract rules such as “hide all private information.” Users instead select regions, correct labels, and state whether similar content should be protected or ignored in the future.

DG2: Make model behavior inspectable and correctable. Privacy errors can be subtle, so the system exposes intermediate decisions rather than only a final edited image. The interface moves from a profile-level overview to region selection, per-region rationales, and edit previews, with direct manipulation at every step.

DG3: Convert local feedback into reusable behavior. A single correction should fix the image at hand and also teach a durable preference, such as “this logo reveals my workplace” or “background strangers are okay,” stored in the Privacy Profile for later runs.

3.2 Two Stages, Two Modes

As Fig. 1 shows, each image moves through two stages that share one interaction pattern: the system proposes, the user corrects, and accepted feedback updates the Privacy Profile, which conditions both stages. **Stage A (Inspect & Select)** proposes privacy-relevant concepts—such as faces, addresses, logos, documents, or background people—and localizes them as regions; the user keeps, deletes, redraws, or adds regions, saying *why* each time. **Stage B (Reason & Generate)** decides whether each selected region is

privacy-relevant in the current context, writes a rationale for it, proposes a mitigation (mask, pixelate, generative replacement, or leave unchanged), and applies the chosen edits for the user to rate.

The workflow runs in two modes. In *Teach* mode, every accepted correction updates the Privacy Profile, so later proposals reflect what the user has taught (RQ2). In *Evaluate* mode, the current profile is applied to unseen images read-only, probing whether learned preferences transfer (RQ1).

3.3 The Privacy Profile

The Privacy Profile is the central learned artifact. Learned preferences can be rendered into natural language for model calls, but a flat text dump would be hard to overview, compare across sessions, or correct when behavior drifts. The profile is therefore an inspectable tree (Fig. 2) in which broad risk categories branch into learned content of three kinds. The categories (identity exposure, location exposure, confidential information, bystander risk, and self-disclosure) follow Imago Obscura’s taxonomy [16].

Concepts are reusable ideas the user has taught, such as “my face,” “my home address,” or “background strangers are okay,” each carrying a severity level and a preferred edit policy. *Examples* ground each concept in confirmed, rejected, redrawn, or safe-marked image regions; positives show what to protect, hard negatives show what must not be confused with the concept, and safe examples become explicit non-risks. *Edit preferences* record how to act, and treat “leave unchanged” as a learned preference rather than the absence of a rule, so the profile represents both risks and contextual exceptions.

The profile is not a weight update but a structured, versioned artifact that can be inspected, rendered to natural language, and exported, giving lightweight provenance over what the user taught and when.

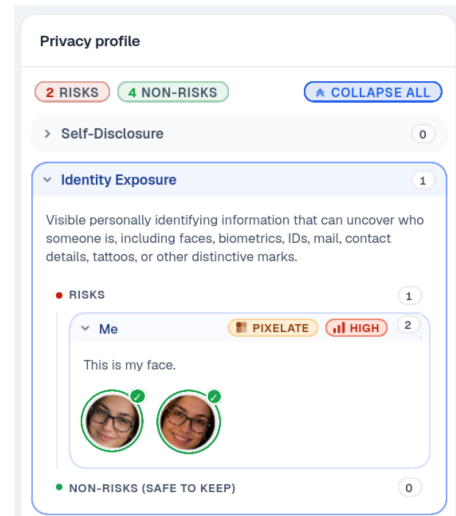


Figure 2: Privacy Profile hierarchy: risk categories branch into learned concepts with severity levels and edit policies, each grounded by example regions.

3.4 Teaching, and the Role of the “Whys”

On the first pass, Stage A starts from a generic privacy taxonomy; learned concepts then guide later proposals. Each correction maps to a profile update: keeping a region adds a positive example, redrawing teaches more precise localization, deleting a mistaken region creates a hard negative, marking a region safe creates a non-risk with a no-op policy, and adding a missed region tells future runs to actively search for similar content. In Stage B, accepting



Figure 3: Stage B in Pringl: proposed policies with rationales (left), the segmented image (center), and the Privacy Profile (right) shown together while the user reviews why each region is sensitive and selects a mitigation.

or overriding a proposed action (replace this face rather than blur it; mask this logo because it reveals location) becomes an edit preference on the corresponding concept. Figure 5 (appendix) walks through this flow for a single region.

User corrections and system proposals both carry a *why*. On correction, the user states the rationale; on proposal, the system writes one from the profile for the user to accept or edit. The text records *which audience matters* and *which harm is feared* in ways a region label cannot. Users can reject or fix a rationale before any edit is applied, which catches misalignment early. The rationales also accumulate into a per-person trace of privacy reasoning that researchers can study directly (Sec. 6).

3.5 Evaluating Transfer to Unseen Images

In Evaluate mode the same two stages run on unseen images, but the profile is applied conservatively and never updated. Regions matching known concepts inherit their learned edit policies; regions matching known non-risks are left unchanged; plausible risks outside the trained profile surface only as low-confidence findings. Scoring the system’s per-region calls against the person’s own labels turns evaluation into a classification problem: true positives are risks caught, false negatives are privacy leaks, and false positives are needless anonymization. This separation makes failures informative. A miss, a wrong region, or a wrong operation indicates a specific gap in the profile (a missing concept, insufficient examples, an unclear non-risk boundary, or a policy that does not transfer), which the user can repair by returning to Teach mode.

4 PRINGL: IMPLEMENTING THE WORKFLOW

Pringl instantiates the pipeline as a web prototype with a staged Next.js frontend and a FastAPI backend that stores sessions, images, annotations, generated outputs, evaluation runs, and versioned Privacy Profile snapshots. One screen combines the image canvas, region list, Privacy Profile, and edit previews (Fig. 3), so users can check a proposed edit against the profile before accepting it.

All model calls use one provider interface. Stage A and Stage B run on a Gemini multimodal model, including helper tasks such as interpreting redraws and dismissals [8]; prompted concepts are segmented with SAM3 [4]. New regions are matched to profile concepts with CLIP for objects, text, logos, documents, and scenes

[22], a face-verification model for identity-specific faces, and Gemini Embedding for labels and descriptions [12]. Generative replacement calls FLUX Kontext when that edit policy is enabled [3].

Model outputs are not shown to the user without backend checks. Deterministic backend checks route regions to face or generic matching, apply confidence thresholds, separate risks from non-risks, and gate Evaluate-mode edits against the current profile, closing the standard interactive-ML loop of data, model, visualization, feedback, and updated output. The source code is available at <https://gitlab.inf.ethz.ch/ou-ivia-courses/xaiml/2026/projects/RuntimeRabbit>, whose README provides step-by-step setup instructions, dependencies, the required Python version, and each member’s contribution; a hosted demo runs at <https://runtimerabbit.xaiml26.ivia.isginf.ch/> at the time of writing.

5 QUALITATIVE RESULTS

We have not yet run a formal user study (Sec. 7); instead we report three Evaluate-mode runs after teaching (Sec. 3.5).

On unseen images, the learned profile applies several taught concepts in one pass (Fig. 4 a–d). In a home-office scene, the door number, wall art, laptop screen, and the person in frame are all anonymized; in a car selfie, the visible home address and the driver are both anonymized.

In a restaurant scene (Fig. 4 e, f), faces are anonymized well, but the system misses the restaurant name on the menus even though it appears on the storefront signage. In our framing this is a false negative, a privacy leak. The location-exposure concept was grounded in signage examples only, so the fix is more teaching, not retraining.

6 DISCUSSION

6.1 Stakeholders and the “Whys”

Users need a system whose anonymization matches their judgment; developers need evidence that automated edits respect individual intent; security researchers need labeled failure cases. The Privacy Profile and its whys serve all three: for users they make the learned model auditable (DG2); for researchers the correction traces record which contexts flip a judgment and which harms people name—raw material for studying privacy reasoning beyond fixed taxonomies.



Figure 4: Evaluate-mode outcomes on unseen images. (a, b) Home office: door number, wall art, laptop screen, and the person in frame are all anonymized. (c, d) Car selfie: the visible home address and the driver are both anonymized. (e, f) Restaurant: faces are anonymized, but the restaurant name on the menus is missed, a false negative because the profile lacked menu examples for that privacy concept.

6.2 Benchmark potential

Because Evaluate mode scores per-region calls against a person’s own labels (Sec. 3.5), profiles and labeled evaluation images double as test cases. Collected at scale, they could become a person-level benchmark, complementing population-level preference learning [11] with person-level alignment evaluation.

6.3 Other domains

The same loop could apply beyond privacy. For example, a platform moderator could build a reusable visual-moderation profile from corrections on flagged uploads.

7 LIMITATIONS AND FUTURE WORK

Scope of privacy claims. Pringl specifies no formal adversary model or guarantee: against a machine-aided adversary with auxiliary data, visually plausible edits may still leave exploitable cues [18, 14]. Embedded text on signs, screens, and documents is also handled inconsistently by region-based models, as Fig. 4 (f) shows; pairing the workflow with formal threat models and text-PII analysis (e.g., Microsoft Presidio [15]) is future work. Our claim is limited to support for articulating and revising preferences, not robustness against strong attacks.

Evaluation. Our evidence is qualitative author sessions, not quantitative re-identification risk, utility loss, or profile-transfer accuracy, and the classification framing of Sec. 3.5 has not yet been run at scale. Our next step is a user study to map individual privacy reasoning more closely and to probe alignment empirically; session data is already stored, but a researcher-facing view for comparing teaching patterns and profile evolution across participants is still missing.

Preference capture and interface cues. The system learns from labels, corrections, whys, and edit choices, but not every “why” behind a judgment is captured—for example, audience-specific intent or unstated trade-offs—and Stage A/B proposals lack calibrated confidence cues that might reduce over-trust before export.

Deployment. The prototype relies on user-supplied API keys and hosted models, so images leave the device during inference despite the “Local” design goal, and sequential stage calls add latency that interrupts iterative teaching. On-device models, caching, and parallel execution are natural next steps.

8 CONCLUSION

We presented a mixed-initiative, co-adaptive workflow and its implementation, Pringl, for teaching context- and user-dependent image privacy through region-level corrections, rationales, and a reusable, inspectable Privacy Profile. Qualitative runs suggest that read-only evaluation on unseen images tests whether a learned profile matches a person’s preferences (RQ1), and that staged teaching lets users correct system behavior over repeated examples (RQ2). A formal user study could supply the labeled sessions needed for a shared benchmark that makes those tests repeatable across models.

9 AI DECLARATION

We used AI tools such as Claude Code (Claude Sonnet 4, Opus 4.8), OpenAI ChatGPT (GPT 5.5), and Google Gemini (Gemini 2.5 Pro) during this project. The problem framing, system goals, privacy rationale, architectural decisions, and evaluation design were defined and executed by us. AI tools supported literature exploration, comparison of implementation options, and feedback on design and writing decisions, as well as smaller utility tasks (e.g., $\LaTeX$ formatting, plotting scripts, and iterative report edits). For implementation and report drafting, we developed the initial structure and core content ourselves and used AI assistance to iterate on specific parts under our direction; when AI outputs were used, we specified constraints, reviewed the results critically, and integrated, revised, or rewrote the final material ourselves. We did not use AI tools to fabricate results or manipulate evidence. The central ideas, argumentation, and conclusions in this report are our own.

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A SUPPLEMENTARY FIGURES

Figure 5 supplements Sec. 3.4 with a single-region teaching walk-through, and Fig. 6 shows an end-to-end view of how one Privacy Profile drives multiple per-region decisions in a single scene.

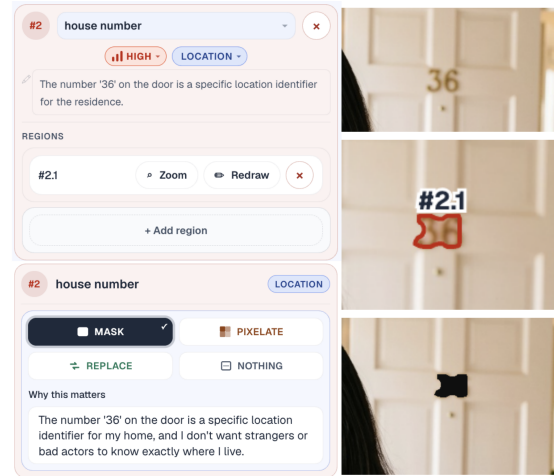


Figure 5: One region from Stage A to Stage B: the system localizes a house number, explains why it matters (“a specific location identifier for the residence”), the user selects a mask action, and the edit is previewed.

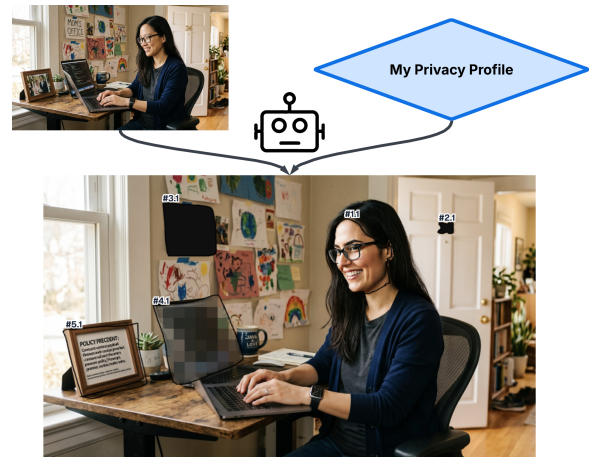


Figure 6: End-to-end processing of one scene: the Privacy Profile guides Stage A localization and Stage B policy application across multiple regions: face, door number, wall art, laptop screen, and a replaced desk photo.